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How tsunami occurs and spreads

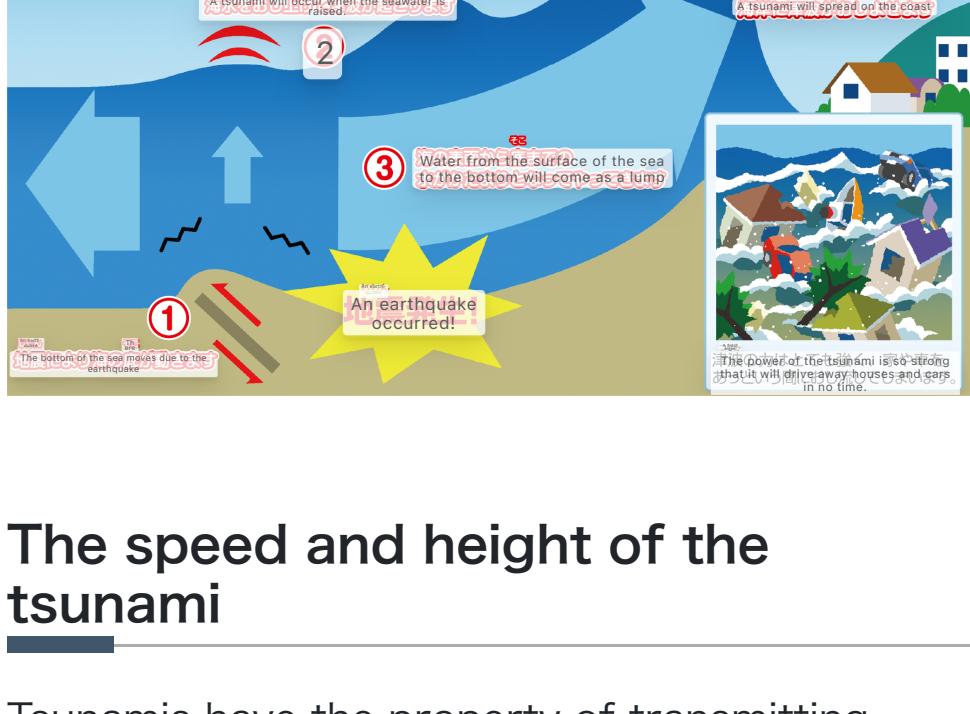
How tsunami occurs and spreads

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Occurrence of tsunami

When a large earthquake occurs under the seabed, the seabed rises or sinks due to fault movement. As a result, the sea level fluctuates, and a tsunami is a large wave that propagates in all directions.

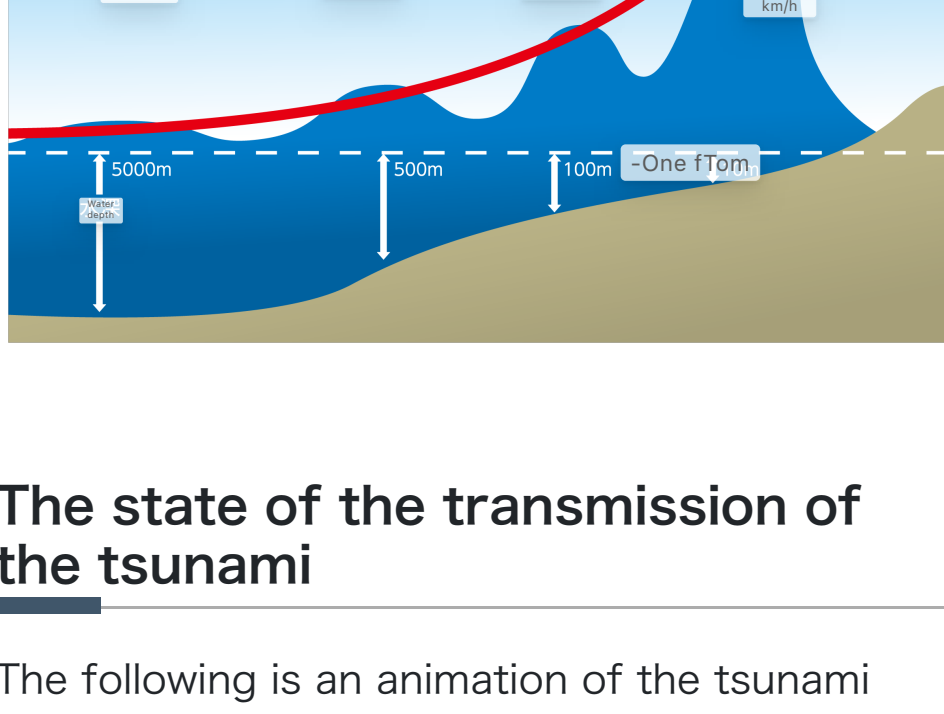
There is a saying that "the tide always recedes before the tsunami", but this is not necessarily the case. Depending on the slope and direction of the underground fault that caused the earthquake, and depending on the location between the place where the tsunami occurred and the coast, large waves may first rush to the coast without the tide receding. **Tsunamis do not necessarily start with a pull wave.**



The speed and height of the tsunami

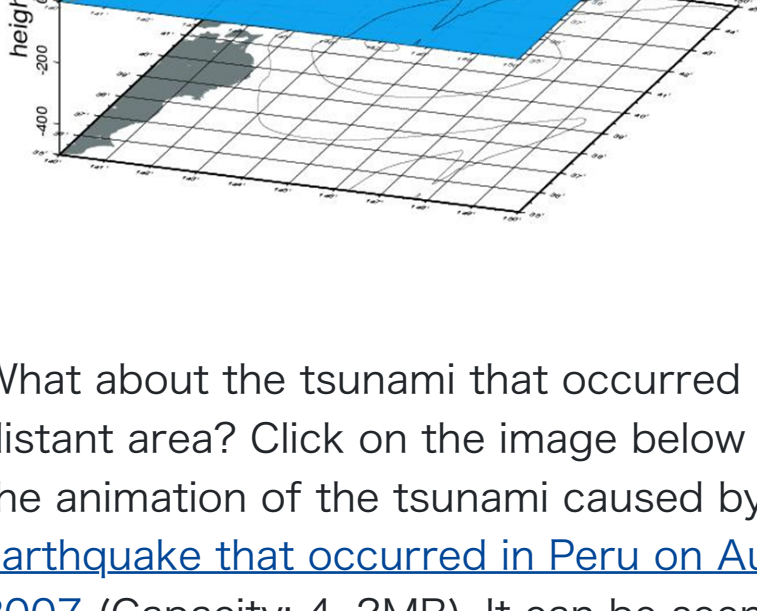
Tsunamis have the property of transmitting faster the deeper the sea, and they are transmitted offshore at a speed comparable to that of a jet plane. Conversely, the shallower the water depth, the slower the speed, so as the tsunami approaches the land, the rear part catches up with the front of the slowed-down wave, and the wave height increases.

Even if it slows down in a shallow water depth, it is not possible for people to run and escape. In order to save your life from a tsunami, it is not in time to start evacuating after seeing the tsunami coming to the coast. If you feel the shaking of the earthquake near the coast, or if a tsunami warning is issued, evacuate quickly even if you can't actually see the tsunami.

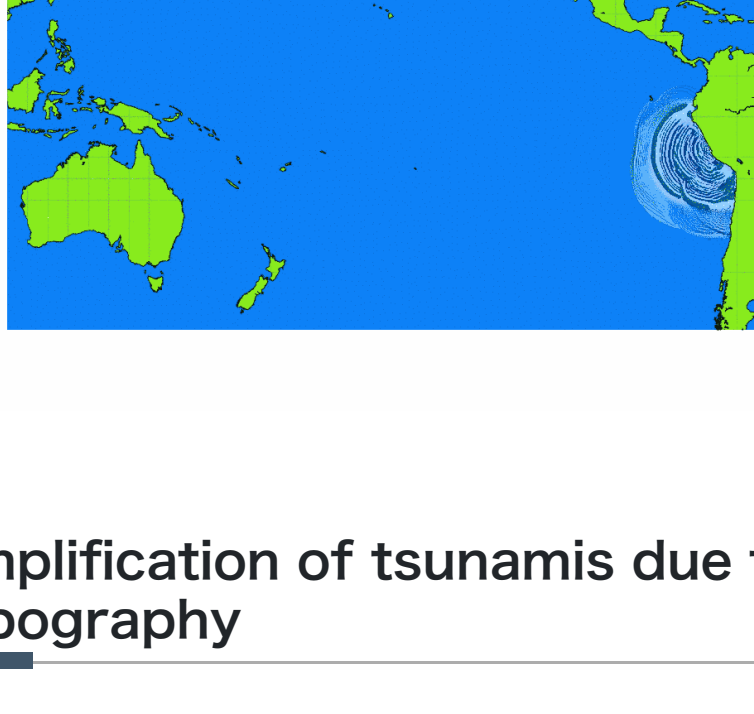


The state of the transmission of the tsunami

The following is an animation of the tsunami caused by the Tokachi coast earthquake in 2003. Click on the image to see the animation (Capacity: 2.7MB). The tsunami that occurred can be seen to be rapidly increasing near the coast.



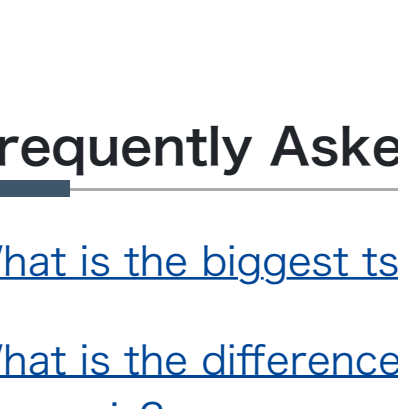
What about the tsunami that occurred in a distant area? Click on the image below to see the animation of the tsunami caused by [the earthquake that occurred in Peru on August 16, 2007](#) (Capacity: 4.2MB). It can be seen that it took more than 20 hours to reach Japan across the Pacific Ocean from Peru, which is far from Japan. The waves are changing in a complex way, while repeating reflections and scattering on the coasts along the way, such as Hawaii. Even after the first wave reaches, it can be seen that subsequent waves are attacking Japan one after another.



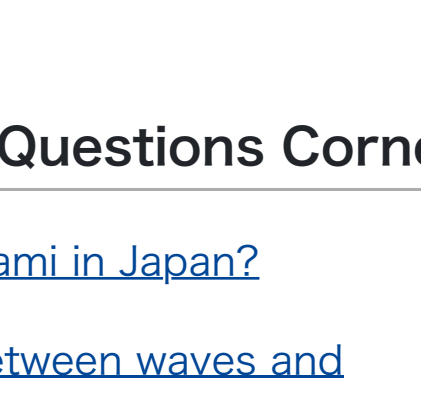
Amplification of tsunamis due to topography

The height of the tsunami varies greatly depending on the terrain near the coast. In addition, tsunamis may run up the land (retrograde). In special terrain locations such as the tip of the cape and the back of the V-shaped bay, the waves are concentrated, so special attention is required. Tsunamis can come in many times by repeating reflections, or multiple waves can overlap and become significantly high waves. For this reason, the first wave is not necessarily the largest, and the tsunami that hits later may be higher.

地形による津波の増幅の例



V字型の湾では湾の奥にエネルギーが集中し、波高が高くなります。



岬の先端では、津波が海岸線に対して平行になろうとエネルギーが集中し、波高が高くなります。

Frequently Asked Questions Corner

[What is the biggest tsunami in Japan?](#)

[What is the difference between waves and tsunamis?](#)

[Why is it dangerous even with a 30cm tsunami?](#)

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